



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Computational Fluid Dynamics

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

Fritz Ali Agildere

January 23, 2026

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Fundamental Definitions

Basic Equations

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Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

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$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + \mathbf{u} \cdot \nabla Q$$

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- *Without Prior Knowledge*: out of n independent measurements the outcome A occurs k times, defining

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Bayes' Theorem

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

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■ *Mean* or first raw moment: $\mu = E[X] = \int_{-\infty}^{+\infty} x f(x) dx$

■ *Variance* or second raw moment: $\sigma^2 = E[(X - E[X])^2] = \int_{-\infty}^{+\infty} (x - E[x])^2 f(x) dx$

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Thank You!

- C. Clarke & B. Carswell, *Principles of Astrophysical Fluid Dynamics* (Cambridge University Press) 2007
- L. A. Barba, *CFD Python: 12 Steps to Navier-Stokes*¹ 2022
- P. Mocz, *Medium*² 2020-2025

Resources on GitHub: [here](#)